

The Constants of Nature

$E_{\rm h}:\text{K}=\frac{\text{kelvin}}{4\pi}\left(\frac{T_p}{m_p}\right)\left(1-14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm de}=\frac{3s}{\mathscr{K}_5\,\mathscr{K}_4}\left(\frac{16}{4\pi}\right)\left(m_{\rm n}+m_{\rm +}\right)\left(1+\frac{Im(\omega_{\rm i})}{\mathscr{K}_2}\left(\frac{L_2}{14}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$\mathscr{E}_{\rm h}=\frac{\text{joule}}{\mathscr{K}_5}\left(\frac{t_p\,q_{\rm p}}{t_p^2}\right)\left(1-\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm h}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_p\,q_{\rm p}}{t_p\,m_e}\right)\left(1+\sqrt{\frac{4}{D_{\rm D0}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$c=\left(\frac{L_2}{T_p}\right)\left(1+\sqrt{G_{\rm G1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$F_{90}=\text{farad}\left(1+\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$K_j=\frac{\mathscr{K}_1}{\pi}\left(\frac{t_p\,q_{\rm p}}{t_p^2\,m_p}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$g_{\rm de}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{t_p^2}\,(\mathscr{K}_1+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\text{K}:E_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_1^4}\left(\frac{m_{\rm p}}{t_p\,m_e}\right)\left(1+14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm u}=\frac{1}{\mathscr{K}_3\,\mathscr{K}_4}\sqrt{\frac{16}{4\pi}}\,(m_{\rm n}+m_{\rm +})\left(1+\left(\frac{16}{2\pi}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$R_{\rm K}=\frac{2\pi}{\mathscr{K}_1^2}\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_p\,q_{\rm p}}{t_p\,m_{\rm +}}\right)\left(1+\sqrt{\frac{4}{D_{\rm D0}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_p}\right)\left(1+\sqrt{G_{\rm G1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$H_{90}=\text{henry}\left(1-\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$\frac{e}{h}=\mathscr{K}_1\left(\frac{t_p\,q_{\rm p}}{t_p^2\,m_p}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$\frac{\mu_{\rm de}}{8\text{K}}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{t_p^2}\,(\mathscr{K}_1+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\Delta_{\text{vcc}}=\frac{E_1}{2\pi}\left(\frac{t_p}{t_p^2\,m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$m_{\rm s}=\frac{-2\pi\mathscr{K}_1}{\mathscr{K}_3}\left(\frac{\text{GeV}}{\sqrt{5}}\right)\left(\frac{t_p^2}{t_p^2}\right)\left(1+\frac{1}{6^{1/2}}\left(\frac{2\pi}{2s}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$Z_0=4\pi\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$r_e=\mathscr{K}_1^2\left(\frac{t_p}{m_e}\right)\left(1-\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$\Omega_{90}=\text{ohm}\left(1-\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\mathscr{K}_1}\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8\text{K}}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{t_p^2}\,(\mathscr{K}_1+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_p}{t_p^2\,m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\,\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_p^2}{t_p^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\,\mathscr{K}_2}\left(\frac{s}{18}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$Z_{\rm p}=\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathscr{K}_1^2}\left(\frac{t_p\,m_{\rm p}}{m_e}\right)\left(1-\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_p^2\,T_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\,\mathscr{K}_2}\right)\,(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_p\,T_{\rm p}}\right)\left(1+\frac{1}{16}\left(\frac{2\pi}{s}\right)^2\,(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2)\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_p}{t_p^2\,m_p}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\,\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_p^2}{t_p^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\,\mathscr{K}_2}\left(\frac{s}{18}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_p}{t_p}\right)\left(1+\frac{1}{2}Im\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm s}^{'}}{S}=\frac{2\,\mathscr{K}_1\,\mathscr{K}_2^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\sigma_{\rm p}=\sqrt{\frac{32}{18}}\,2\pi\mathscr{K}_1^4\left(\frac{t_p\,m_{\rm p}}{t_p^2}\right)^2\left(1-2\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_p^2\,T_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\,\mathscr{K}_2}\right)\,(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4)\boxtimes\right)$	$\frac{h}{e}=2\pi\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_p^2\,m_{\rm p}}{t_p\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_p\,T_{\rm p}}\right)\left(1+\frac{1}{16}\left(\frac{2\pi}{s}\right)^2\,(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2)\boxtimes\right)$
$\text{Hz}:\text{J}=\frac{2\pi}{\text{second}}\left(\frac{t_p^2\,m_{\rm p}}{t_p}\right)\left(1+\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$\frac{G}{hc}=\text{GeV}^2\left(\frac{t_p^2}{t_p^2\,m_{\rm p}}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$A_{\rm vel}=\mathscr{K}_1^2\left(\frac{t_p}{t_p}\right)\left(1-\frac{1}{2}Im\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm s}^{'}}{\mu_{\rm B}}=\frac{2\,\mathscr{K}_1\,\mathscr{K}_2^2}{S}\left(\frac$				

$t_p = 5.39125836832313 \times 10^{-40}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-38}$ m	Planck length
$q_p = 1.87554596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \times 10^{-7}$	hyperbolic universe boundary
5	number of dimensions
7	break in scale symmetry
$\pi_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\pi_2 = 3.66756753480501 \dots$	2 nd hyperbolic partition constant
$\pi_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\pi_4 = 1.877649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\pi_5 = 4.4876244916751 \dots$	hyperbolic radius constant
$\pi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{00} = 1.01494160640965 \dots$	Gieseking's constant
$V_k = 0.209598321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.915965554772190 \dots$	Catalan's constant
$C_d = 0.2916506090430818 \dots$	domino tiling constant
$D_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649747018528596 \dots + 1.3249706214087 \dots i$	omega_1 constant
$\omega_2 = 1.5295940370597192 \dots$	omega_2 constant
$i^{i^{i^i}}$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$\phi_i = (-1)^{1/2}$	imaginary golden ratio
$G_9 = 1.51872847301812 \dots$	tether length for grazing half the unit circle
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$S = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.244115108584261 \dots$	arc length of the unit lemniscate
$L = 2.62205755429219 \dots$	lemniscate constant
$L_1 = 1.311028771462015 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.884713084793979 \dots$	ubiquitous constant
$G_{66} = 0.834626841674073 \dots$	Gauss's constant
$C_{CF} = 0.697774657964819 \dots$	continued fraction constant
$R_{FR} = 0.998136044598059 \dots$	Ramanujan's 1 st continued fraction constant
$F = 1.22642102702035 \dots$	Fibonacci factorial constant
$M = -1.477456459463318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.82282549678847 \dots$	Sierpinski's constant
$C_{JF} = 2.4048255670595772 \dots$	1 st root of the Bessel function
$\alpha_{\text{beta}} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{90} = 0.739085133215160 \dots$	Dottie number
$L_0 = 0.662743419349181 \dots$	Laplace limit
$a_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\delta = 4.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.246479603371714 \dots$	silver constant
$p = 1.32471795724474 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$q_{QA} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{\text{GSM}} = 0.3532363781854995 \dots$	Hafner-Sarnak-McCurley constant
$T_0 = 2.731500000000000 \dots \times 10^5$ kelvin	freezing point of water
$p = 1.000000000000000 \dots \times 10^2$ pascal	1 bar
$P_1 = 1.01325003754773 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules	energy associated with Δv_{CS}
$f_1 = 5.400000000000000 \dots \times 10^{14}$ Hz	frequency of max light wavelength
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 3.24149397212207 \dots$	peak emission of spectral flux/unit frequency
$\log(x)$ = natural logarithm	
$\Gamma(x)$ = gamma function	ln = lumen = cd · sr
$\zeta(x)$ = Riemann zeta function	N = newton = m kg/s ²
ln = derangement function	N = noether = m kg/s
	O = ohm = m ² kg/s ² C ²
units	P = alpha / kg/s ² m
m = meter	S = siemens = s C ² /m ² kg
m = second	T = tesla = kg/s ² C
C = coulomb	V = volt = m ² kg/s ³ C
K = kelvin	W = watt = m ² kg/s ³
kg = kilogram	Wh = weber = m ² kg/s · C
derived units	
MHz = megahertz = 1.000000000000000 ... × 10 ⁶ 1/s	
fm = femtometer = 1.000000000000000 ... × 10 ⁻¹⁵ m	
eV = electron-volt = 1.000000000000000 ... eV	
MeV = megaelectron-volt = 1.000000000000000 ... × 10 ⁶ eV	
GeV = gigaelectron-volt = 1.000000000000000 ... × 10 ⁹ eV	
$r_e = \frac{4\pi}{(r_s)^2}$	$r_s = \sqrt{\frac{C_v}{4}}$
$r_r = \frac{r_n}{r_s}$	$r_o = \sqrt{\frac{2\pi}{\log(36)}}$
$\epsilon_p = \frac{l_p^2}{r_p^2} \frac{m_p}{r_p}$	
external transform space	Physis Monastery
$V_{24} = \left(\frac{4\pi}{5t} \right) \left(\frac{4\pi}{15} \right) \left(\frac{4\pi}{35} \right) \left(\frac{4\pi}{18} \right) \left(\frac{4\pi}{32} \right) \left(\frac{4\pi}{8} \right)^4$	Thad Roberts
figure eight knot hyperbolic volume	October-28-2025
$V_0 = \left(\left\{ L_2 \left(\frac{1}{\phi_1} \right) - L_2(x) \right\} \right)$	with:
binomial constructor	Angela Arvizu
$A_{\text{ext}} B_{\text{ext}} (1 + A_{\text{int}} \Xi)$	Vanessa Monz
$\Xi = \frac{l_p m_p}{\rho_p^2} \times \frac{\text{coulomb$	